

TASK 1 - OSI LAYERS

LAYER 1 - PHYSICAL LAYER

Definition

The Physical Layer is Layer 1 of the OSI Model. It is responsible for transmitting raw bits (0s and 1s) through a physical medium.

Function

- Transmits bits as physical signals.
- Defines physical transmission characteristics.
- Handles physical media, connectors, and signaling.
- Defines characteristics such as transmission speed.

Physical Media

- Copper cables
- Fiber-optic cables
- Wireless radio
- Coaxial cables

Examples

- Ethernet cable
- Fiber-optic cable
- RJ45 connector
- Wireless radio signals

Data Unit : Bits

Cybersecurity Relevance

- Physical access can enable unauthorized network access.
- Protect network equipment and cables.
- Physical security is part of overall network security.

Key Points

- Layer 1 = Physical Layer
- Deals with raw bits
- Uses physical signals
- Includes cables, connectors, and transmission media
- Data unit = bits
- MAC addresses belong to Layer 2, not Layer 1

Quick Revision

Layer 1 physically moves bits from one device to another using signals over a transmission media.

Remember

For a wired Ethernet connection:

PC → Bits → Electrical signals → Ethernet cable → Switch

For fiber:

PC → Bits → Light pulses → Fiber-optic cable → Switch

For Wi-Fi:

Device → Bits → Radio signals → Air → Access Point

LAYER 2 - DATA LINK LAYER

Definition

The Data Link Layer is Layer 2 of the OSI Model. It provides local network communication and organizes bits into frames.

Function

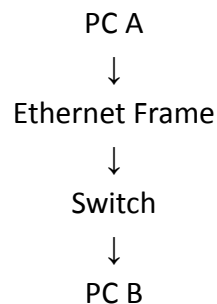
- Creates and handles frames.
- Uses MAC addresses for local delivery.
- Provides communication between devices on the same local network.
- Performs error detection at the link layer.

Data Unit: Frame

Addressing: MAC Address

Common Device: Switch

Example



Key Points

- Layer 2 = Data Link
- PDU = Frame
- Uses MAC addresses
- Switches primarily operate at Layer 2
- Handles local network communication
- Layer 2 sits above Physical Layer

Security Relevance

- MAC addresses can help identify network devices.
- Layer 2 attacks include ARP spoofing/poisoning.
- Controls include port security, VLANs and 802.1X.

Quick Revision

Layer 2 takes bits from Layer 1, organizes them into frames, and uses MAC addresses to help deliver those frames across a local network.

LAYER 3 - NETWORK LAYER

Definition

The Network Layer is Layer 3 of the OSI Model. It provides logical addressing and routing of packets between networks.

Function

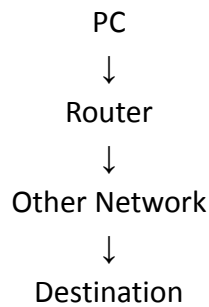
- Provides logical addressing using IP addresses.
- Determines how packets are forwarded between networks.
- Performs routing.
- Connects different networks.

Data Unit: Packet

Addressing: IP Address

Common Device: Router

Example



Layer 2 vs Layer 3

Layer	Data Unit	Address	Main Function	Device
Layer 2	Frame	MAC	Local delivery	Switch
Layer 3	Packet	IP	Routing between networks	Router

Security Relevance

- SOC analysts monitor source and destination IPs.
- IP addresses help investigate network connections.
- Suspicious external IP communication may indicate malicious activity
- Routers and Layer 3 controls help segment and route networks.

Key Point

Layer 3 uses IP addresses to route packets between different networks.

LAYER 4 - TRANSPORT LAYER

Definition

The Transport Layer is Layer 4 of the OSI Model. It provides end-to-end communication between applications/processes on networked devices.

Main Functions

- Provides application/process-to-process communication.
- Uses port numbers to identify transport endpoints.
- Performs segmentation and reassembly.
- TCP provides reliable, ordered delivery.
- TCP manages connections.
- UDP provides connectionless, lightweight communication.

Main Protocols

Protocol	Meaning	Characteristics	Data Unit
TCP	Transmission Control Protocol	Connection-oriented, reliable, ordered	Segment
UDP	User Datagram Protocol	Connectionless, lightweight	Datagram

Important Terms

Port:

Identifies a transport-layer communication endpoint/service.

TCP:

Provides reliable, connection-oriented communication

UDP:

Provides connectionless communication with lower overhead.

Examples:

IP Address -> 192.168.1.10

Port -> 443

Protocol -> TCP

Cybersecurity Relevance

- SOC analysts monitor source/destination ports.
- Port information helps identify network services.
- Unusual connection patterns may indicate scanning or malicious activity.
- TCP behavior can be useful during network investigations.

Key Point

Layer 4 provides end-to-end communication between applications using transport protocols such as TCP and UDP, with port numbers identifying communication endpoints.

LAYER 5 - SESSION LAYER

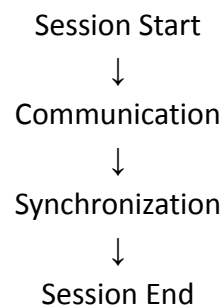
Definition

The Session Layer is Layer 5 of the OSI Model. It establishes, manages, synchronizes, and terminates communication sessions between applications.

Functions

- Establishes sessions
- Maintains sessions
- Synchronizes communication
- Terminates sessions

Example



Key Points

- Layer 5 = Session Layer
- Manages communication sessions
- Handles session establishment and termination
- Provides synchronization/checkpoints

Quick Revision

Layer 5 manages the ongoing communication session between applications.

Synchronization: Provides coordination/checkpoints during an ongoing session.

LAYER 6 - PRESENTATION LAYER

Definition

The Presentation Layer is Layer 6 of the OSI Model. It handles data representation and transformation so information can be correctly understood by communicating systems.

Main Functions

- Data Translation
- Encoding and Decoding
- Encryption and Decryption
- Compression and Decompression

Examples

- ASCII
- Unicode / UTF-8
- Encryption
- Compression

HTTPS Connection

HTTPS uses TLS to provide cryptographic protection for HTTP communication.

Key Points

- Layer 6 = Presentation Layer
- Handles data representation
- Can handle encoding and decoding
- Associated with encryption/decryption
- Can handle compression/decompression

Quick Revision

Layer 6 transforms data into a suitable format and can provide functions such as encryption and compression

Encoding is not encryption. Encoding changes data representation for compatibility; encryption protects data using cryptographic techniques.

LAYER 7 - APPLICATION LAYER

Definition

The Application Layer is Layer 7 of the OSI Model and provides network services and protocols used by applications.

Main Functions

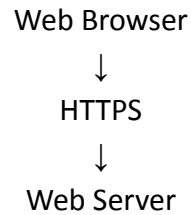
- Provides network services to applications.
- Enables applications to communicate over networks.
- Defines application-level network protocols.

Example

- HTTP
- HTTPS
- DNS
- DHCP

- FTP

Real-World Example



Key Points

- Layer 7 = Application Layer
- Closest to the end user
- Uses application-layer protocols
- Examples: HTTP, HTTPS, DNS, DHCP, FTP

Quick Revision

Layer 7 provides network services and protocols that applications use to communicate across a network.

ONE LINE MEMORY

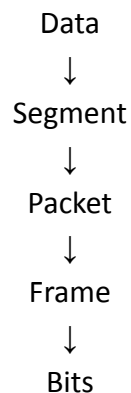
Application -> Presentation -> Session -> Transport -> Network -> Data Link -> Physical

ENCAPSULATION & DECAPSULATION

Encapsulation

Definition

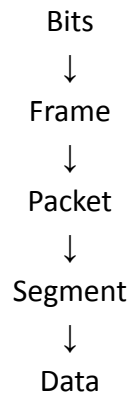
The process of adding layer-specific information as data moves down the OSI layers on the sender.



Decapsulation

Definition

The process of removing/processing layer-specific information as data moves up the OSI layers on the receiver.



PDU Names

OSI Layer	PDU (Protocol Data Unit)
Layer 7-5	Data
Layer 4	Segment (TCP)/Datagram (UDP)
Layer 3	Packet
Layer 2	Frame
Layer 1	Bits

Key Points

- Encapsulation = down the stack
- Decapsulation = up the stack
- Each layer adds/processes information for its function.
- Sender encapsulates
- Receiver decapsulates

Quick Revision

Sender : Data -> Segment -> Packet -> Frame -> Bits

Receiver : Bits -> Frame -> Packet -> Segment -> Data

TASK 2 - TCP/IP MODEL

Definition

The TCP/IP model is a 4-layer model that describes network communication using the TCP/IP protocol suite.

Four Layers

4. Application
3. Transport
2. Internet
1. Network Access

Functions

Application

- Provides network services to applications.
- Examples: HTTP, HTTPS, FTP, DNS, DHCP.

Transport

- End-to-end communication.
- Uses TCP/UDP
- Uses port numbers.

Internet

- Logical addressing and routing
- Uses IP
- Data unit: Packet

Network Access

- Local network and physical transmission
- Includes MAC, frames, and physical media.

OSI vs TCP/IP Mapping

OSI 7 + 6 + 5 -> TCP/IP Application
OSI 4 -> TCP/IP Transport
OSI 3 -> TCP/IP Internet
OSI 2 + 1 -> TCP/IP Network Access

Key Point

TCP/IP has 4 layers, while OSI has 7 layers; TCP/IP combines several OSI layers.

TCP/IP MODEL - IMPORTANT PROTOCOLS

Application Layer

- HTTP -> Web Communication -> TCP 80
- HTTPS -> Secure web communication -> TCP 443
- FTP -> File transfer -> TCP 21
- DNS -> Domain name resolution -> commonly UDP 53; can also use TCP
- DHCP -> Automatic network configuration -> UDP 67/68

Transport Layer

- TCP -> Connection-oriented, reliable, ordered
- UDP -> Connectionless, lightweight

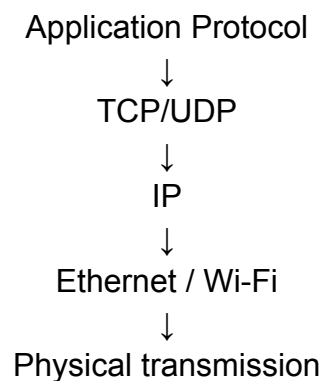
Internet Layer

- IP -> Logical addressing and routing

Network Access Layer

- Ethernet
- Wi-Fi
- MAC addresses
- Frames
- Physical transmission

Key Flow



Key Point

Protocols at different TCP/IP layers work together to provide complete network communication.

COMPARISON

OSI vs TCP/IP Model

Definition

- 7 layers
- Detailed reference model
- Separates Application, Presentation, and Session
- Separates Data Link and Physical

TCP/IP Model

- 4 layers
- Represents TCP/IP protocol architecture
- Combines OSI Application + Presentation + Session
- Combines OSI Data Link + Physical

Layer Mapping

OSI 7 + 6 + 5 -> TCP/IP Application

OSI 4 -> TCP/IP Transport

OSI 3 -> TCP/IP Internet

OSI 2 + 1 -> TCP / IP Network Access

Key Differences

OSI = 7-layer reference model

TCP/IP = 4-layer model based on the TCP/IP protocol suite

Quick Revision

OSI = 7 -> 6 -> 5 -> 4 -> 3 -> 2 -> 1

TCP / IP = 4 -> 3 -> 2 -> 1

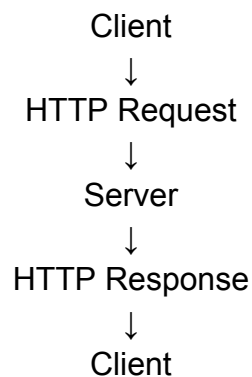
TASK 3 - PROTOCOLS

HTTP - Hypertext Transfer Protocol

Definition

HTTP is an Application Layer Protocol used for communication between clients and servers , especially for web resources.

How It Works



Port - TCP 80

Common Methods

- GET -> Request/retrieve resource
- POST -> Send data to server

Status Code Categories

- 2xx -> Success
- 3xx -> Redirection
- 4xx -> Client/request error
- 5xx -> Server error

Security Concerns

- HTTP does not provide TLS-based cryptographic protection.
- Sensitive information sent over unprotected HTTP may be exposed to interception

Key Points

- HTTP = Hypertext Transfer Protocol
- Application Layer protocol
- Application Layer protocol
- Uses request-response communication
- Common default port = TCP 80
- HTTPS is the protected form of HTTP using TLS

Quick Revision

HTTP allows a client and server to communicate using requests and responses, commonly over TCP port 80.

HTTP uses a request-response model and commonly uses TCP port 80. Traditional HTTP does not provide TLS-based cryptographic protection.

HTTPS - Hypertext Transfer Protocol Secure

Definition

HTTPS is HTTP communication protected using TLS (Transport Layer Security).

Function

- Provides secure web communication.
- Protects data in transit
- Provides confidentiality.
- Provides integrity protection.
- TLS can authenticate the server.

Port - TCP 443

Relationship -> HTTP + TLS = HTTPS

HTTP vs HTTPS

HTTPS protects sensitive HTTP communication from unauthorized observation or tampering while in transit.

Key Point

HTTPS is HTTP protected by TLS, commonly using TCP port 443

FTP - File Transfer Protocol

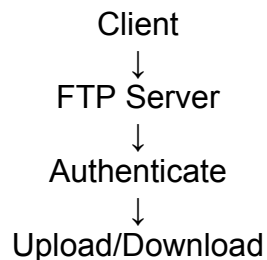
Definition

FTP is an Application Layer Protocol used to transfer files between a client and server over a network.

Function

- Upload files
- Download files
- Manage file transfers between client and server

How It works



Port = TCP 21 (FTP Control Connection)

FTP also uses a separate data connection for transferring files.

Security Concerns

- Traditional FTP does not provide TLS protection.
- Credentials may be exposed.
- File contents may be exposed to network observers.
- Secure alternatives include FTPS and SFTP.

Cybersecurity Relevance

- Monitor FTP connections.
- Investigate unexpected FTP usage.
- Be cautious with sensitive files transferred using encrypted FTP.

Key Point

FTP transfers files between clients and servers, traditionally using TCP port 21 for control communication, but traditional FTP does not provide TLS encryption.

FTP uses TCP; port 21 is used for the control connection. Traditional FTP does not provide TLS protection. FTPS uses TLS, while SFTP operates over SSH.

DNS - Domain Name System

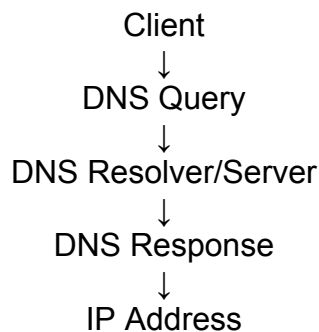
Definition

DNS is an Application Layer system/protocol that resolves domain names into IP addresses.

Function

- Resolves domain names to IP addresses.
- Allows users to use human-readable names instead of IP addresses.
- Uses a query-response process.

How It Works



Port 53

- Commonly UDP 53
- Can also use TCP 53

Important Record

A Record -> Domain name -> IPv4 address

Security Concerns

- DNS Spoofing
- DNS tunneling
- Suspicious/malicious domain queries

Cybersecurity Relevance

DNS activity can help identify suspicious domains and unusual network behavior.

Key Point

DNS translates domain names into IP addresses so network communication can reach the correct destination.

DNS queries are normal network activity. Security concerns arise when DNS behavior is unusual, such as suspicious domains, abnormal query volumes, or DNS tunneling.

DHCP - Dynamic Host Configuration Protocol

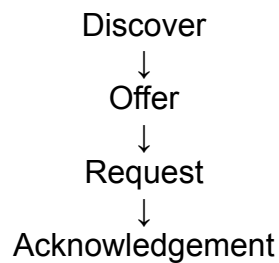
Definition

DHCP is an Application Layer Protocol that automatically provides network configuration to devices.

Provides

- IP Address
- Subnet mask
- Default gateway
- DNS server information

DHCP Process - DORA



Ports

- UDP 67 - Server
- UDP 68 - Client

Security Concerns

- Rogue/unauthorized DHCP servers
- Incorrect or malicious network configuration
- Unexpected gateway/DNS settings

Key Point

DHCP automatically provides network configuration to devices using UDP ports 67 and 68.

TASK 4 - IP Addressing

IP Addressing

Definition

An IP address is a logical address used to identify a network interface/device for communication using IP.

IPv4

- IPv4 uses 32 bits.
- Written as four decimal octets.
- Each octet contains 8 bits.
- Each octet ranges from 0-255.

Example: 192.168.1.10

Key Points

- IP address = logical addressing
- IP operates at OSI Layer 2
- IPv4 = 32 bits
- 4 octets
- Each octet = 8 bits
- Each octet = 0-255

MAC vs IP

MAC -> Layer 2 -> local/link addressing

IP -> Layer 3 -> logical addressing/routing

Quick Revision

An IPv4 address is a 32-bit logical address written as four 8-bit decimal octets.

Public vs Private

Private IP

Used inside private/local networks.

Private IPv4 ranges

10.0.0.0/8

172.16.0.0/12

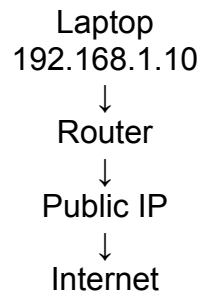
192.168.0.0/16

Example: 192.168.1.10

Public IP

Used for communication over the public internet.

Example



Key Points

- Private IPs are used inside local/private networks.
- Public IPs are used for Internet-facing communication.
- 192.168.1.10 is private.
- Private IPv4 ranges are not directly routable on the public internet.
- Routers commonly use NAT to connect private networks to the Internet.

Quick Revision

Private IP = Internal network

Public IP = Public network

Static vs Dynamic IP

Static IP

An IP address that is manually/configurationally assigned and normally remains the same.

Dynamic IP

An IP address assigned automatically, usually by DHCP, and may change over time.

Comparison

Static IP	Dynamic IP
Manually assigned	Usually assigned by DHCP
Predictable	May change
Normally stays the same	Automatic Configuration

Key Point

Static IPs are predictable and normally remain fixed; dynamic IPs are automatically assigned and may change.

A dynamic IP may change depending on DHCP lease/configuration; disconnecting Wi-Fi does not automatically guarantee a new IP.

Loopback Address

Definition

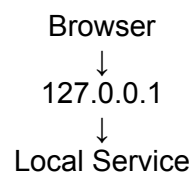
A loopback address is an IP address used by a device to communicate with itself.

IPv4 Loopback

127.0.0.1

Hostname = localhost

Example



Key Points

- 127.0.0.1 is the most commonly used IPv4 loopback address.
- Localhost commonly refers to the local machine.
- Loopback communication stays within the host.
- Useful for testing local network applications/services.

Quick Revision

127.0.0.1 (localhost) means “this computer” and also this is a IPv4 loopback address used for communication with the local host.

Subnet Mask

Definition

A subnet mask determines which part of an IPv4 address represents the network portion and which part represents the host portion.

Example

IP address: 192.168.1.10
Subnet Mask: 255.255.255.0

192.168.1 | 10
Network Host

Binary

255.255.255.0 = 11111111.11111111.11111111.00000000

Therefore: 24 network bits, 8 host bits

CIDR -> 255.255.255.0 = /24

Key Points

- Subnet mask separates network and host portions.
- 255.255.255.0 is /24
- /24 has 24 network bits and 8 host bits.
- Devices with the same network portion can be in the same subnet.
- Different subnets generally require routing through a router.

Quick Revision

A subnet mask tells a device which part of an IPv4 address represents the network and which part represents the host.

CIDR - Classless Inter-Domain Routing

Definition

CIDR is a notation that represents an IP address together with the number of network bits.

Example

192.168.1.10/24

- IP address = 192.168.1.10
- /24 = 24 network bits
- Host bits = $32 - 24 = 8$

CIDR ↔ Subnet Mask

/24 = 255.255.255.0

/16 = 255.255.0.0

/8 = 255.0.0.0

Key Points

- CIDR uses /number notation.
- The number represents network bits.
- IPv4 has 32 total bits.
- Host bits = $32 - \text{prefix length}$.

Quick Revision

CIDR /24 means the first 24 bits are network bits and the remaining 8 bits are host bits.

Basic Subnetting

Definition

Subnetting is the process of dividing a larger IP network into smaller subnetworks.

Main Idea

Borrow host bits to create additional network bits and smaller subnets.

Example

Original : 192.168.1.0/24

Subnet into : 192.168.1.0/26

/26

26 network bits , 6 host bits

Number of Host Addresses

$2^6 = 64$ addresses per subnet

Usable Host Addresses

$64 - 2 = 62$ usable hosts

Four /26 Subnets

192.168.1.0/26
192.168.1.64/26
192.168.1.128/26
192.168.1.192/26

First Subnet

Network : 192.168.1.0
Hosts : 192.168.1.1 - 192.168.1.62
Broadcast : 192.168.1.63

Key Points

- Subnetting divides a network into smaller networks.
- Borrowing host bits creates more subnets.
- /26 has 26 network bits and 6 host bits.
- /26 provides 64 total addresses and normally 62 usable host addresses.
- Network and broadcast addresses are not normally assigned to hosts.

Quick Revision

Subnetting borrows host bits to create smaller networks.

Finding a subnet

For a /26:

Block size = 64

Subnet boundaries :

0, 64, 128, 192

For an IP address:

1. Find the range containing the IP.
2. First address = network address.
3. Last address = broadcast address.
4. Addresses between them = usable host addresses.

